Answer **all** the questions in spaces provided.

1. Genetically modified plasmids were the first vectors developed and are still widely used for cloning of genes. Such plasmids are often used to introduce foreign genes into bacterial cells and these recombinant bacteria will in turn be used either in the mass production of useful proteins or in the transformation of cells to form genetically modified organisms.

(a) What are plasmids?

Small/not part of bacterial chromosome/extrachromosomal, [1]

circular/loop DNA molecules found naturally in bacteria;

@ 1mk. If 1 or 2 correct, award ½ mk

(b) State three essential features of a plasmid for it to be used as a vector for the cloning of genes in bacteria.

Able to replicate itself and any DNA fragment it carries/has site/origin of replication;

Contain at least one recognition sequences/restriction site (to allow insertion of DNA fragments to be cloned);

It can gain entry into a host cell / will not be degraded in host cell;

Contains (at least 2) marker gene(s) (that reveal their presence in host cells/help in selection);

[3]

(c) Using a **named** example, explain the advantages of being able to transfer human genes into bacteria.

E.g. insulin / human growth hormone / bovine somatotrophin;

Cheap / fast method of production;

of large quantities of the human protein;

Identical to those produced in human body / that will not stimulate immune response;
Overcomes ethical objections/does not kill animals (vs. production from live animals);
Eliminates risk of zoonosis; [3]
Reduces the need to acquire proteins from other sources; any 3 @ 1

(d) Suggest why plasmids by themselves cannot be used to insert genes directly into plant cells.

Plasmids cannot enter plant cells through the cellulose cell wall;

Plant cells do not have competent factor for plasmid uptake;

Plasmids can only passed between bacteria;

(In general) Plasmids are not integrated into host genome/ will be degraded; [1]

any 1 @1 mark,

There is a concern that transgenic organisms pose a threat should they be released into the environment. One method of detecting such organisms is to introduce a further gene that codes for an enzyme, **luciferase**, along with the desired gene. This enzyme occurs naturally in a number of marine bacteria and causes bioluminescence (the production of light) in the presence of oxygen and the substrate luciferin. The reaction that takes place is shown below.

luciferin + ATP + oxygen → oxyluciferin + AMP + light

(e) What is meant by a ***transgenic organism***?

Organism with a foreign gene integrated into its genome (ref. to foreign is impt);
Award ½ mk for genetically/artificially modified.

The foreign gene is expressed/product is formed by organism/ ref. to new trait; [2]

(f) Describe the main steps involved in producing colonies of transformed *E coli* with the luciferase gene.

Isolate plasmids (from *E. coli)* and luciferase gene;

They are cut separately; by same restriction enzyme;

Cut plasmids and genes are mixed; sticky ends anneal;

with DNA ligase added (formation of phosphodiester bonds);

Product added to/inserted into (competent) bacterial cells; via electroporation/heat shock;

Bacteria plated on agar; containing luciferin;

Bioluminescent colonies (on agar plate) are selected; [4]

(g) Suggest an advantage of using the luciferase gene to detect the presence of escaped genetically modified organisms. [1]

Easy to detect as genetically modified organisms produce light;

Provide quantitative results; Safe/non-toxic method of detection;

Does not require removal of cells from organisms;

Cheap/ref. to cost of technique; @1 mark, any 1 [1]

[Total: 15 marks]

2. In order to study embryonic stem cells, scientists extract the inner cell mass from blastocysts and culture the cells *in vitro*. The embryonic stem cells must be grown in a special culture medium that maintains the pluripotency status of the cells. Without the appropriate growth factors, the embryonic stem cells will spontaneously differentiate.

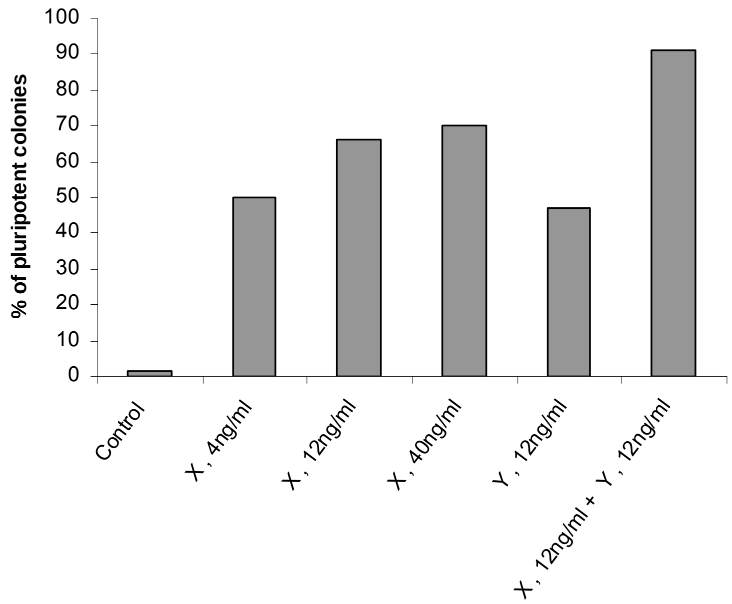
(a) Telomerase is not usually active in cells of multicellular organisms. With reference to a feature of stem cells, suggest whether embryonic stem cells possess active telomerase.

Yes they possess telomerase;

Because they are capable of self-renewal/ ref. to cell division; [2]

Two growth factors, **X** and **Y**, were added to embryonic stem cell culture to study their effect on the pluripotency status of the stem cells. Pluripotency status was measured by the presence of a pluripotency marker called Oct-4 on the cultured cells (Fig. 2.1).

Fig. 1.1 Effect of growth factors X and Y on the pluripotency of cultured embryonic stem cells



(b) (i) With reference to Fig. 2.1, describe the effect of **X** and **Y** on the pluripotency of cultured embryonic stem cells.

1. X increases % of pluripotent colonies;

as conc. of X increases from 4-40ng/ml OR 0-40ng/ml, % of pluripotent colonies increases from 50-70% OR [1 to 2] -70%;

2. Y also increases the % of pluripotent colonies; as conc. of Y increases from 0-12ng/ml, % of pluripotent colonies increases from [1 to 2] - [45 to <50] %;

3. X has greater effect than Y at same conc. ;
at 12ng/ml, [65 to <70]% vs [45 to <50]% ;

4. X and Y together will give rise to highest /optimum % OR Effect of X + Y greater than X or Y alone ; of (>90 to 95)%;

as compared to X alone(at 12ng/ml), [65 to <70]% of pluripotent colonies OR
Y alone (at 12ng/ml), [45 to <50]%;

5. At higher conc of X, increase in conc brings about a smaller increase in % of pluripotent colonies (than at lower conc of X) / vice versa;

When X increases from 0-4ng/ml, % increases from ~1 - 50% OR
when X increases from 4-12ng/ml, % increases from 50 - [65 to <70]%;

whereas when X increases from 12-40ng/ml, % increases from [65 to <70] - 70%;
@½mk [3]

(ii) Both growth factors are protein molecules which are not able to enter the embryonic stem cell. Suggest how the growth factors are still able to cause the cells to remain pluripotent.

Growth factors bind to a receptor (idea of) on cell membrane/surface of cell;

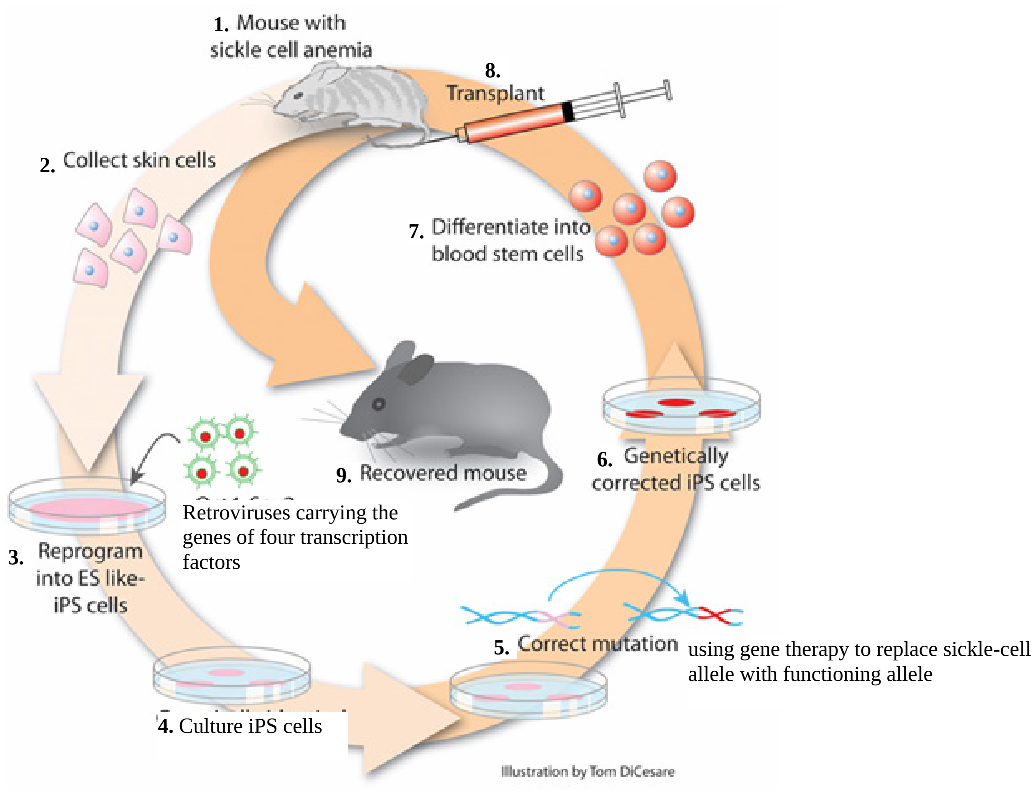
Activate receptors and so (idea of) activate second messengers/signal transduction pathway;

Ref. to a response / specific eg. activate enzymes/protein kinases/ cause cell division/turn on/off specific genes; @ 1mk [2]

(c) The use of embryonic stem cells is ethically controversial as embryos are destroyed in the process. As such, scientists have devised another way of obtaining pluripotent stem cells.

Differentiated mouse skin cells were 'reprogrammed' by infection with retroviruses carrying the genes for four different transcription factors. With the incorporation of these four genes, the cells became pluripotent and showed many characteristics of embryonic stem cells. These cells were termed induced pluripotent stem (iPS) cells.

Fig. 1.2. shows how scientists have cured mice with sickle-cell anaemia. iPS cells were generated from a sick mouse (1-4). The sickle-cell allele in the iPS cells was replaced using gene therapy (5-6). The iPS cells with the functional allele were made to differentiate into blood stem cells (7) before transplantation back into the same mouse, which cured the mouse (9).



(i) State the function of transcription factors.

Control rate of transcription / initiate/ promote transcription /help recruit RNA polymerase / forms TIC;

by binding to control elements/promoter; @ ½ mk [1]

(ii) Suggest how the insertion of genes coding for transcription factors can cause a differentiated cell to become pluripotent. [2]

Ref. to genes code for/ directing synthesis of transcription factors;

which will then activate/inhibit (idea of) specific genes;

proteins formed enable cell to be pluripotent/ ref. to divide continuous/ability to differentiate; @ 1 mk [2]

(iii) Describe three social and ethical considerations for the use of gene therapy. [3]

Social: **Religious belief - (concerns that scientists are overstepping their bounds by) manipulating organisms via transferring genes across natural boundaries is seen as usurping the prerogative/right of God;**

Cost – issue of who should pay for gene therapy with ref. to 1 eg. insurers/ taxpayers/government;

Issue of whether it is acceptable to use gene therapy for enhancement/aesthetics (A! ref. to named eg. height, intelligence, athletic ability)** ****can be ethical too**;**

Ethical: the appropriate time that gene therapy should be used with ref. to 1 eg. babies/children, the critically ill; [3]

the issue of defining what is a disability/disorder/disease (A! ref. to named eg. dwarfism, intelligence, obesity);

The unknown effects that would germline gene therapy have on future generations;

Deciding who should have access to gene therapy / ref. to access only by those who can afford it;

@ 1mk, any 3. Combination of 3 must include at least 1 social and 1 ethical consideration

(d) Unlike animals, all cells of a plant are totipotent. This characteristic has been exploited in producing genetically-modified crops as one genetically-modified plant cell is able to divide and differentiate to form a whole plant.
Describe, with named examples, **three** improvements in quality and yield of crop plants that have been achieved using genetic engineering.

Herbicide resistant plants such as Roundup Ready/ resistant to glyphosate/Roundup;

Famers can use herbicide to effective control weeds / ref. to increase yield since weeds are controlled;

Insect-resistant plants such as Bt plants (A! Potato leafroll virus-resistant potato / Papaya ringspot virus-resistant papaya);

ref. to increase yield since pests are destroyed / less pesticides need to be used on crops;

<u>Flavr Savr</u> tomato;
Longer shelf-life/ stay fresher longer/ improved taste;

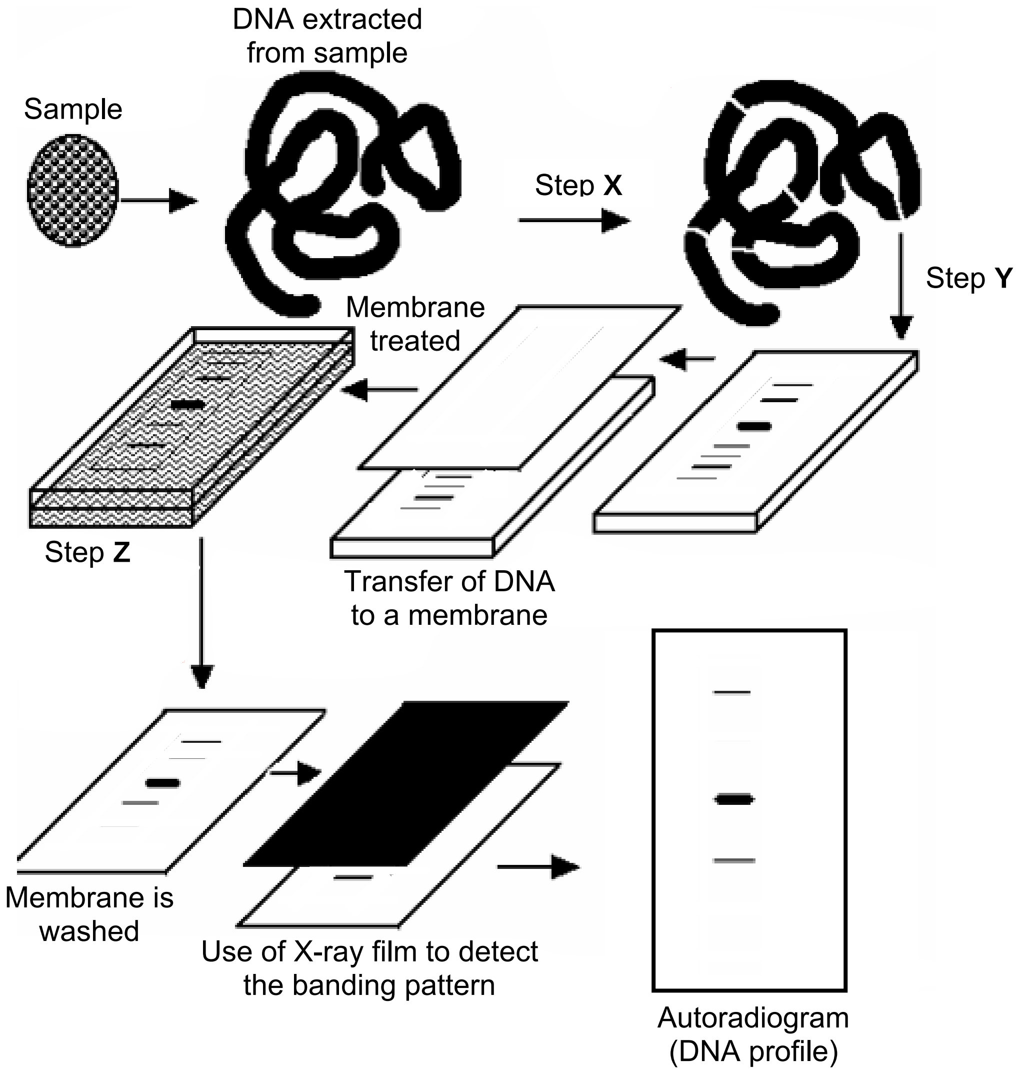
<u>Golden Rice</u>;

Increased levels of carotene (R! vit. A) / increase vit. A amounts in humans / increase nutritional value @ ½ mk

A! edible vaccines against eg. hepatitis B, diarrhea, cholera and measles;
cheap source of vaccine; @ ½ mk [3]

[Total: 16 marks]

3. The figure below shows the process of Southern blotting using a sample.



a) i) Briefly explain the following steps:

Step X **Digest/Cut with restriction enzymes; at specific restriction sites;**

To produce (restriction) fragments;

[2]

Step Y **(Gel) electrophoresis; DNA, being negatively charged, moves towards positive end; shorter fragments traveling the furthest;**

[2]

Step Z **Soak in radioactive (single-stranded nucleic acid) probe; sequence complementary to sequence/gene of interest;**

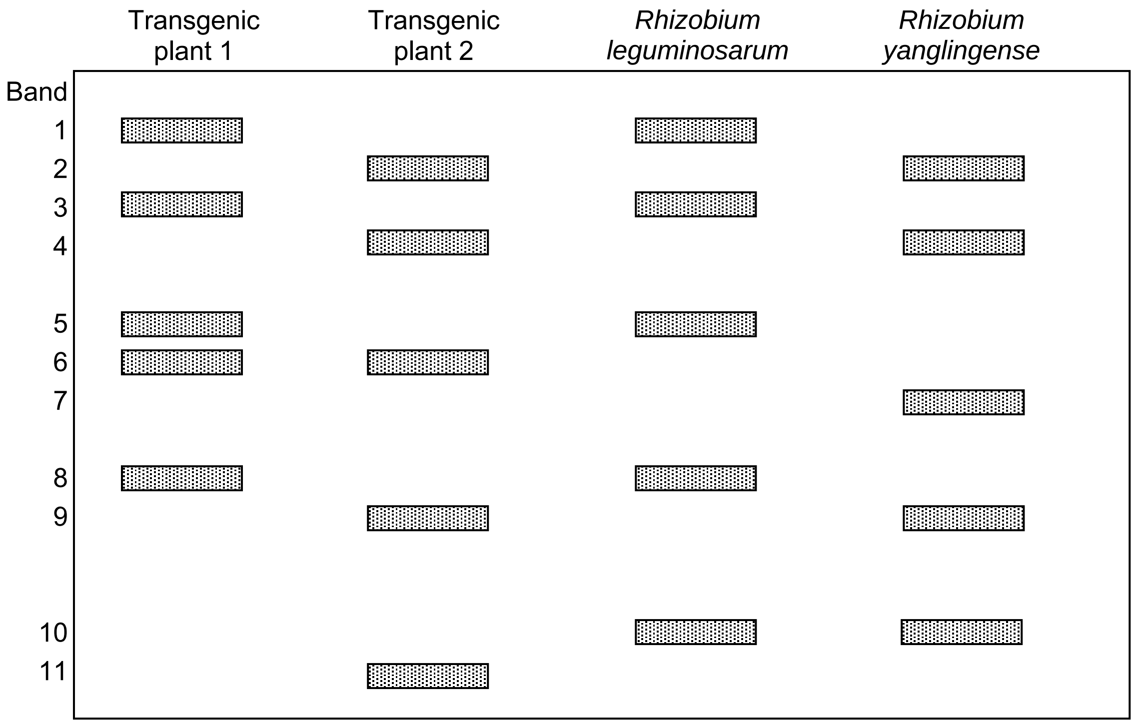
[2]

ii) Suggest a reason for the need to produce the autoradiogram for analysis rather than using the results obtained after Step Y.

Ref. to DNA not visible to the naked eye / more permanent record /

autoradiogram shows a subset, or smaller number of bands for easier analysis; [1]

The figure below shows the DNA profiles of two transgenic plants and a couple of bacteria that are found in the soil around these plants.



b) i) Describe the process by which bacteria are able to obtain DNA from their surroundings.

Transformation; Ref. to competent cells / competence factor; producing (membrane) proteins that facilitate uptake of DNA ; @ ½ mk

ii) Based on the figure above, match the bacteria to the transgenic plant that it is likely to be found with. Explain your answers.

***R. leguminosarum* found near plant 1 and *R. yanglingense* found near plant 2;**

First pair: common bands of 1, 3, 5, 8 and 2nd pair: common bands 2, 4, 9;

Award ½ if only common bands of one pair given.

DNA/genes from plant may have entered the soil bacteria;

[3]

iii) Suggest one implication of genetically modified plants shown by the DNA profiles above.

Transfer of antibiotic/herbicide resistance/foreign genes in transgenic plants to

bacteria / other organisms; [1]

iv) Suggest why the quality of transgenic plants grown from seeds produced by existing transgenic plants may differ from those grown by tissue culture.

Genome of seeds may not be identical to parents / genome of plantlets from tissue culture are identical to parents ;

due to crossing over and independent assortment during meiosis (A! either crossing over or independent assortment alone) ;

favourable traits may be mixed with unfavourable ones; [2]

[Total: 14 marks]

Essay Question:

(a) With a named example, explain how RFLP analysis facilitates the process of
(i) disease detection [4]

1. **Process allows the detection of change in sequence of nucleotide of a DNA molecule;**
2. **Due to inability of the restriction enzyme to cut the DNA at the particular site;**
3. **Resulting in a different mixture of fragments from each DNA molecule;**
4. **Sorted by gel electrophoresis / nucleic acid hybridization/Southern blotting;**
5. **And different band patterns for the different DNA molecules/alleles;**
6. **Used for identification of disease-causing alleles/mutations/organisms;**

@ 1 mk, max 4 mks

(ii) genomic mapping [2]

1. **RFLPs are used as genetic markers; R! gene/genome markers, marker genes**
2. **To construct <u>linkage</u> maps;**
3. **and to determine the order of the restriction fragments/genetic markers / to determine the relative distances between genetic markers along a chromosome (based on recombination frequencies);**

(b) What are the benefits that can be derived from sequencing the human genome? [7]

A. Molecular medicine (no marks for heading. Max 2 mks, @ 1 mk)

1 Earlier diagnosis/detection of genetic diseases;
2 Gene therapy;
3 Rational drug design / control systems for drugs / rational drug design / pharmacogenomics 'custom drugs' ;

B. Energy and Environmental Applications (max 1 mk, @ 1 mk)

4 Use microbial genomics research to create new energy sources (biofuels);
5 Use microbial genomics research to develop environmental monitoring techniques to detect pollutants ;
6 Use microbial genomics research for safe, efficient environmental remediation;

C. DNA Forensics (max 3 mk, @ 1 mk)

7 Identify potential suspects whose DNA may match evidence left at crime scenes;
8 Exonerate persons wrongly accused of crimes;
9 Identify crime and catastrophe victims;
10 Establish paternity and other family relationships;
11 Identify endangered and protected species as an aid to wildlife officials (could be used for prosecuting poachers);
12 Detect bacteria and other organisms that may pollute air, water, soil, and food;
13 Match organ donors with recipients in transplant programs;
14 Determine pedigree for seed or livestock breeds;
15 Authenticate consumables such as caviar and wine;

D. Agriculture, Livestock Breeding, and Bioprocessing (max 1 mk, @ 1 mk)

16 Healthier, more productive, disease-resistant crops/ farm animals / higher yield;
17 More nutritious produce ;
18 Edible vaccines incorporated into food products;
19 New environmental cleanup uses for plants like tobacco;

E. Bioarchaeology, anthropology, evolution and human migration (max 1 *mk, @ 1 mk*)

20 Study human evolution (through germline mutations in lineages);
21 Study of migration of diff pop groups based on female genetic inheritance/lineage and migration of males via Y chromosomes;
22 Compare breakpoints in the evolution of mutations with ages of populations and historical events;

F. Risk assessment (@ *1 mk*)
23 Assess health damage and risks caused by radiation exposure/mutagenic chemicals/cancer-causing toxins;
Total max: 7 mk

(c) Discuss briefly the ethical issues raised by the human genome project. [7]

1. **Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others / who should have access;**

2. **Privacy and confidentiality of genetic information / who owns, who can control;**

3. **Possible discrimination for individuals with known genetic defects/prone to genetic diseases in different areas of life / abortion of fetuses with minor disorders;**

4. **Psychological impact and stigmatization due to an individual's genetic differences;**

5. **Reproductive issues such as the adequacy of information for parents/individuals and the rights for decision making for complex and potentially controversial procedures;**

6. **The reliability of the genetic tests for pre-natal decision making, eg. abortion.**

7. **Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions.**

8. **Conceptual and philosophical implications regard human responsibility/degree of genetic influence on behaviour and hence the party responsible/the line between medical disease and enhancement;**

9. **Health and environmental issues concerning genetically modified foods (GM) and microbes.**

10. **Property rights/Commercialization of products (patents, copyrights, and trade secrets)**

11. **Fairness in the accessibility of data/material/technologies.**

Any of the above points, max 7